Expert-Guided Software Team Formation in Practice: An Optimization Tool, Practitioner Validation, and Lessons Learned [Supplementary Material]

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I. Introduction

This document accompanies *Expert-Guided Software Team Formation in Practice: An Optimization Tool, Practitioner Validation, and Lessons Learned*. It supplies detailed experimental evidence and distinguishes selected-team quality from run-average quality and the runtime budget used to obtain recommendations.

Table S1 characterizes the industrial dataset used to calibrate and evaluate the pipeline, including the candidate pool, technology taxonomy, temporal project split, team-size distribution, and collaboration graph. Table S2 documents the retained practitioner-validation design and distinguishes unique cases, repeated comparisons used to examine consistency, and isolated acceptability judgments.

Table S3 compares MILP and GA quality on twelve controlled size-four profiles; its time ratio uses the mean of one GA run, not the cost of the best-of-thirty recommendation. Table S9 provides both GA budgets. Table S4 compares calibrated-GA recommendations with historical teams on sixteen reconstructed 2025 project instances. Their P-identifiers do not denote the controlled RQ2 profiles.

Unless explicitly labeled V2, the detailed numerical tables below concern the original V1 evaluation. They allow readers to trace its aggregate results to project-level evidence. Section III separately summarizes the refined evaluation and identifies its different evidence units; V1 scores must not be read as V2 results.

Table S5 documents GA calibration. Table S6 gives the selected-team AE and AC of the greedy and random baselines; Table S7 characterizes the five uniform-preferred teams in the separate practitioner evaluation. Table S8 compares the selected direct-LLM response with calibrated-GA recommendations.

The computational V2 comparison in Section III-B selects each GA by its own objective and then evaluates the selected teams on the common refined scale. It is separate from the V1 baseline table and from the practitioner's judgments.

The original-evaluation package and the separately versioned V2 baseline workbook provide result records for their respective comparisons. They support inspection and numerical recalculation, not full execution-level reproduction from raw organizational inputs and LLM sessions.

TABLE S1
Summary of the industrial dataset.

Aspect	Summary
Developer inventories	512 records with technology information; excluding two report-only records without graph alignment yields the 510-developer candidate pool and direct-LLM input.
Canonical technology taxonomy	16 domains, 214 ecosystems, and 23 programming languages (253 terms); taxonomy size is distinct from observed terms in an experimental subset. Median per developer: 2 domains, 6 ecosystems, and 3 languages.
Calibration projects	205 projects starting up to 2024. Team sizes: k=4: 45, k=5: 38, k=6: 37, k=7: 30, k=8: 26, k=9: 29.
Held-out projects	16 projects starting in 2025. Team sizes: k=4: 4, k=5: 4, k=6: 1, k=7: 1, k=8: 2, k=9: 4.
Collaboration graph	Consolidated co-participation history transformed through the expert-calibrated compatibility function; not established as an as-of-staffing-time snapshot.
Historical requirements	Reconstructed from historical members' aggregate competence profiles: 100% occurrence = Must; 50–99% = Should; other observed technologies = Could. Applies to RQ3–RQ5.

TABLE S2
RETAINED V1 HISTORICAL/UNIFORM PRACTITIONER-VALIDATION DESIGN. THE SEPARATE UNCONTAMINATED LLM COMPARISONS AND V2 ROUND ARE DISTINGUISHED IN SECTION III.

Task	Unique cases	Repeats	Rows
Calibrated GA vs. historical	16	3	19
Calibrated GA vs. uniform GA	12	2	14
Calibrated-GA acceptability	16	–	16

TABLE S3
RQ2 COMPARISON BETWEEN THE EXACT MILP REFERENCE AND THE CALIBRATED GA ON THE TWELVE SIZE-FOUR CONTROLLED BENCHMARK PROJECTS.

Project	MILP AE	GA best AE	Best gap (%)	GA AE mean ± SD	Hits/30	MILP time (s)	One-run ratio
P1	0.818783	0.818783	0.00	0.788474 ± 0.050642	12	109.69	7.85×
P2	0.844445	0.844445	0.00	0.769284 ± 0.059689	7	208.44	10.26×
P3	0.813492	0.810847	0.33	0.789697 ± 0.032128	0	186.75	7.61×
P4	0.864286	0.864286	0.00	0.817566 ± 0.033847	1	45.74	4.41×
P5	0.843915	0.826720	2.04	0.771497 ± 0.032369	0	34.46	2.20×
P6	0.781746	0.781746	0.00	0.753527 ± 0.035620	14	508.73	93.51×
P7	0.870900	0.859326	1.33	0.858016 ± 0.004189	0	585.89	117.37×
P8	0.857672	0.857672	0.00	0.790728 ± 0.054681	10	195.47	27.33×
P9	0.867725	0.867725	0.00	0.854559 ± 0.023670	16	122.19	15.06×
P10	0.900794	0.900794	0.00	0.876323 ± 0.024483	11	105.33	26.60×
P11	0.834656	0.834656	0.00	0.779374 ± 0.031629	2	73.92	12.84×
P12	0.877513	0.857672	2.26	0.794899 ± 0.028884	0	226.69	45.79×
Summary	–	–	0.50 mean	–	73/360	–	13.95× median

TABLE S4
RQ3 BACKTEST ON THE SIXTEEN HELD-OUT 2025 PROJECTS.

Project	k	$\mathrm{AE}_{\mathrm{real}}$	$\mathrm{AE}_{\mathrm{GA}}$	Gain (%)	$\mathrm{AC}_{\mathrm{real}}$	$\mathrm{AC}_{\mathrm{GA}}$
P1	9	0.5238	0.6114	16.72	0.5143	0.8111
P2	8	0.3750	0.5167	37.78	0.3643	0.6000
P3	5	0.4532	0.6822	50.53	0.4200	0.7200
P4	9	0.4008	0.6506	62.32	0.3571	0.8167
P5	4	0.5762	0.6365	10.47	0.6000	0.8667
P6	7	0.5246	0.5393	2.81	0.7667	0.7952
P7	5	0.5817	0.6326	8.75	0.6333	0.8600
P8	5	0.4365	0.6757	54.80	0.4000	0.8400
P9	6	0.4048	0.5311	31.20	0.4000	0.6467
P10	8	0.5869	0.6162	5.00	0.6643	0.8429
P11	9	0.3274	0.5041	53.98	0.3071	0.6833
P12	4	0.3770	0.5178	37.35	0.3667	0.7667
P13	4	0.7817	0.8643	10.56	0.7667	0.9000
P14	4	0.4008	0.9008	124.75	0.3000	0.9000
P15	5	0.3770	0.5347	41.84	0.3667	0.7400
P16	9	0.4373	0.4963	13.48	0.4810	0.7667
Mean	–	0.4728	0.6194	35.15	0.4818	0.7847

Among the eight controlled benchmark projects where the GA reached the certified MILP optimum, four recovered the identical team up to member ordering (P1, P6, P10, P11), and four reached a different team with the same optimal AE, indicating alternative optima under the same objective (P2, P4, P8, P9).

TABLE S6

PER-PROJECT V1 SELECTED-TEAM AE AND AC FOR THREE ALGORITHMIC BASELINES ON SIXTEEN RECONSTRUCTED 2025 INSTANCES. GREEDY RUNS ONCE; RANDOM AND RANDOM-SEARCH SELECT THE BEST CALIBRATED AE AMONG THIRTY OUTPUTS. ALL COMPONENTS REFER TO THAT SAME SELECTED TEAM. AT IS OMITTED HERE; ITS SELECTED-TEAM MEANS IN THE MAIN PAPER ARE 0.5470 FOR RANDOM AND 0.5641 FOR RANDOM-SEARCH. UNIFORM-GA RESULTS ARE EXCLUDED FROM THIS V1 SUMMARY.

Project	Greedy AE	Greedy AC	Random AE	Random AC	Random-Search AE	Random-Search AC
P1	0.3869	0.3500	0.3908	0.3556	0.5026	0.4889
P2	0.3393	0.3214	0.3683	0.3571	0.4441	0.4429
P3	0.3865	0.3400	0.4190	0.3800	0.6178	0.6400
P4	0.3763	0.3278	0.3988	0.3556	0.5106	0.4889
P5	0.5119	0.5000	0.4563	0.4333	0.6225	0.6667
P6	0.3690	0.3476	0.3843	0.3667	0.4809	0.4810
P7	0.3952	0.3600	0.4119	0.3800	0.5935	0.7200
P8	0.4198	0.3800	0.4196	0.3800	0.6340	0.7200
P9	0.3881	0.3800	0.3992	0.3933	0.4628	0.5000
P10	0.5000	0.4857	0.3933	0.3571	0.5022	0.4929
P11	0.3446	0.3278	0.3677	0.3556	0.4386	0.4889
P12	0.3492	0.3333	0.4294	0.4333	0.4707	0.6667
P13	0.4206	0.3333	0.4778	0.4333	0.6905	0.6667
P14	0.4008	0.3000	0.5079	0.4333	0.7897	0.7667
P15	0.3214	0.3000	0.3891	0.3800	0.4937	0.5400
P16	0.3585	0.3444	0.3668	0.3556	0.4386	0.4889
Mean	0.3918	0.3582	0.4113	0.3844	0.5433	0.5787

TABLE S5

GA CALIBRATION VIA ADAPTIVE RACING.

Aspect	Summary
Chronological split	205 pre-2025 projects: 144 training, 31 validation, 30 test.
Selected configuration	Population 150, 200 generations, elitism 3, mutation rate 0.005, crossover rate 1.0, 20 stable generations, CCC crossover operator.
Baseline configuration	Population 100, 100 generations, elitism 1, mutation rate 0.05, crossover rate 1.0, 10 stable generations, CCC crossover operator.
Test-set comparison	150 paired test-set runs per configuration. Selected mean fitness 0.6120; baseline mean fitness 0.5983; relative gain 2.29%; paired outcomes 108 wins, 35 losses, 7 ties.

TABLE S7

RQ5 BLINDED PAIRWISE COMPARISONS: AT/AC OF THE CALIBRATED AND $GA_{UNIFORM}$ TEAMS FOR THE FIVE PROJECTS WHERE THE PARTICIPANT SELECTED $GA_{UNIFORM}$ OVER THE CALIBRATED GA. IN EVERY CASE THE UNIFORM TEAM HAD HIGHER AC, LOWER AT, AND A LARGER $|AT - AC|$ GAP.

Project	Calibrated AT	Calibrated AC	Uniform AT	Uniform AC
P2	0.5000	0.6000	0.4413	0.7786
P7	0.5872	0.8600	0.5638	0.8800
P8	0.6429	0.8400	0.6354	0.8600
P11	0.4683	0.6833	0.4286	0.8111
P12	0.4680	0.7667	0.4345	0.9000

a) Direct-LLM input sanitization.: The input contained 510 pseudonymized developers aligned with the collaboration graph; two records without graph nodes were excluded. Ninety-one target-project history links were removed from 78 developer records. The graph itself was unchanged, so this control removes specified direct links rather than all possible retrospective information. The public workbook lists source-file metadata, but the detailed sanitization manifest and actual prompt/raw-response files are not included in the inspected release. One complete valid response was retained before scoring; the sixteen project outputs are not sixteen independent LLM executions.

TABLE S8

PER-PROJECT AT, AC, AND AE FOR THE SELECTED DIRECT-LLM RESPONSE AND THE CALIBRATED-GA BEST-OF-THIRTY RECOMMENDATION ON SIXTEEN RECONSTRUCTED 2025 INSTANCES. ALL SIXTEEN LLM TEAMS PASSED CHECKS FOR KNOWN IDENTIFIERS, DISTINCT MEMBERS, EXACT TEAM SIZE, AND FULL MUST COVERAGE.

Project	LLM AT	LLM AC	LLM AE	GA best AT	GA best AC	GA best AE	AE gap (GA − LLM)
P1	0.5714	0.4167	0.4425	0.5714	0.8111	0.6114	0.1689
P2	0.4286	0.4000	0.4048	0.5000	0.6000	0.5167	0.1119
P3	0.6190	0.6200	0.6192	0.6746	0.7200	0.6822	0.0630
P4	0.6190	0.4389	0.4689	0.6174	0.8167	0.6506	0.1817
P5	0.5714	0.7000	0.5929	0.5905	0.8667	0.6365	0.0437
P6	0.4762	0.5381	0.4865	0.4881	0.7952	0.5393	0.0528
P7	0.5714	0.5000	0.5119	0.5872	0.8600	0.6326	0.1207
P8	0.6190	0.5200	0.5365	0.6429	0.8400	0.6757	0.1392
P9	0.4286	0.5533	0.4494	0.5079	0.6467	0.5311	0.0817
P10	0.5714	0.5214	0.5298	0.5709	0.8429	0.6162	0.0864
P11	0.4286	0.4611	0.4340	0.4683	0.6833	0.5041	0.0701
P12	0.4286	0.5667	0.4516	0.4680	0.7667	0.5178	0.0662
P13	0.6905	0.5000	0.5317	0.8571	0.9000	0.8643	0.3326
P14	0.9048	0.5667	0.6230	0.9048	0.9000	0.9008	0.2778
P15	0.4286	0.4000	0.4048	0.4937	0.7400	0.5347	0.1299
P16	0.4286	0.4833	0.4377	0.4422	0.7667	0.4963	0.0586
Mean	0.5491	0.5116	0.4953	0.5866	0.7847	0.6194	0.1241

TABLE S9
RQ2 RUNTIME BUDGET ACCOUNTING. THE QUALITY ATTACHED TO BEST-OF-THIRTY GA SELECTION MUST BE INTERPRETED WITH THE THIRTY-RUN TOTAL, NOT A SINGLE-RUN MEAN. TIMES ARE SECONDS.

Project	MILP recorded	GA mean run	GA thirty-run total	MILP/GA mean	MILP/GA total
P1	109.69	13.975544	419.266317	7.85	0.262
P2	208.44	20.315477	609.464316	10.26	0.342
P3	186.75	24.545726	736.371792	7.61	0.254
P4	45.74	10.363741	310.912233	4.41	0.147
P5	34.46	15.678240	470.347195	2.20	0.073
P6	508.73	5.440133	163.204004	93.51	3.117
P7	585.89	4.991713	149.751385	117.37	3.912
P8	195.47	7.152856	214.585674	27.33	0.911
P9	122.19	8.110933	243.327989	15.06	0.502
P10	105.33	3.960247	118.807412	26.60	0.887
P11	73.92	5.757125	172.713750	12.84	0.428
P12	226.69	4.950352	148.510559	45.79	1.526
Median	154.47	7.631895	228.956832	13.95	0.465

II. RUNTIME BUDGETS AND EVIDENCE PROVENANCE

The same thirty GA runs used for best-of-thirty quality have a cost distinct from one execution. Table S9 uses the recorded runtime fields in the RQ2 result workbook. The thirty-run total is accumulated runtime, not measured parallel latency. The two ratios divide recorded MILP time by, respectively, mean one-run GA time and total thirty-run GA time. Ratios below one favor the recorded MILP time. The table does not establish common hardware conditions or whether the MILP field includes both ordering subproblems; a portable performance comparison requires those details.

The recorded MILP time is lower than the thirty-run total in nine of twelve profiles. The sum of all 360 GA run times is 3,757.262626 seconds. The median of per-project ratios is not the ratio of the two runtime medians. Table S10 summarizes the evidence units and their provenance boundaries.

TABLE S10

EVIDENCE UNITS AND PROVENANCE BOUNDARIES. MODEL LABELS IDENTIFY THE EVALUATED ARTIFACTS RATHER THAN IMPLY INTERCHANGEABLE HISTORICAL VERSIONS OR AN AS-OF-TIME DATA SNAPSHOT.

Evidence	Evaluated model	Inputs and selection unit	Interpretation boundary
Calibration	GA using closed-form AE	205 pre-2025 projects; 144/31/30 chronological split	No 2025 instance used for configuration selection.
RQ1	V1 BN reference and projected surrogate	26,250 enumerated scenarios; 1,200 real-team configurations	Numerical and fixed-team ranking fidelity; no V2 BN validation or complete-search comparison.
RQ2	Same calibrated AE in GA and MILP	Twelve controlled size-four profiles; thirty GA runs each	Separate benchmark from the sixteen historical instances; timing-accounting limitations above.
RQ3–RQ4 V1	V1 calibrated AE for common rescoring	Sixteen roster-reconstructed 2025 instances; consolidated profiles/graph	Excludes the corrected uniform-GA comparison; requirements are not independent as-of-staffing-time inputs.
RQ4 V2	Refined AE for common rescoring	Same sixteen instances; 480 runs per GA, 480 random draws, sixteen greedy outputs	Each GA selects by its own objective; Random-Search excluded; no pooling with V1.
Direct LLM	Interface model label GPT-5.6 Sol; High reasoning mode	August 27, 2026; 510 graph-aligned developers; first valid response	Computational scores; practitioner-evaluation protocol described in Section III.
RQ5 V1	Blinded judgments, no displayed AE	One authority; sixteen historical and twelve distinct uniform comparisons	Five reversed-order repeats are checks, not additional independent cases; separate LLM condition summarized below.
TeamPlus	Fixed-seed calibrated GA	New six-person case; seed 1127947874; twenty ranked candidates	One facilitated task; no historical-roster requirement reconstruction.

III. Refined Evaluation and Evidence Boundaries

V2 modifies the contribution of optional technology coverage, useful mandatory redundancy, and collaboration-score scale. It retains technology-axis weights $(3, 1, 5)$, the AE bottleneck ratio $(5, 1)$, and hard mandatory coverage. Positive coverage weights were recalibrated using the original six elicitation scenarios, not subsequent practitioner choices. The refined formula respects fourteen of fourteen elicited order relations, compared with thirteen before refinement. Collaboration centroids become $(0.3, 0.4, 0.5, 0.6, 0.7)$ through $\mathrm{AC}' = 0.5 + 0.5(\mathrm{AC} - 0.5)$. No reimplemented V2 BN is evaluated here.

A. Fixed-Team and Exact-Reference Checks

The V1 fixed-team fidelity check additionally yields mean Spearman correlation 0.794, pairwise concordance 0.859, top-ten overlap 0.800, and top-twenty overlap 0.775 across six sets of 200 teams. A top team is shared in three of six sets; mean BN-score regret of surrogate selection is 0.0237 and the maximum is 0.0605. These values do not establish complete-search or V2 BN fidelity.

For V2, the twelve size-four MILP instances all reached certified optima with zero gap. Best-of-thirty GA matched nine optima, with mean absolute gap 0.0028, mean relative gap 0.3879%, and maximum gap 2.5986%. All 360 GA runs were feasible. Mean recorded MILP time was 121.0 seconds versus 9.21 seconds for one GA run; best-of-thirty quality requires the full thirty-run budget.

A sensitivity check rescored 960 fixed teams with refined technology coverage and compressed collaboration. Comparing bottleneck with arithmetic-mean aggregation gave mean Spearman correlation 0.9840 but changed the top team in nine of sixteen projects. The mean top-five intersection was 3.31 teams. This is a rescoring analysis, not an optimizer rerun or evidence that one aggregation produces better human judgments.

B. V2 Algorithmic Baselines

The post-refinement comparison uses the same sixteen reconstructed project instances but evaluates every selected team under V2. Refined calibrated GA and uniform GA each execute thirty seeds per project. Each representative maximizes its own search objective, with exact ties resolved by the smallest seed. Uniform search uses simple means; its selected output is only subsequently evaluated using refined AT, AC, and AE. Random draws thirty candidates per project and reports the feasible candidate with highest refined AE; greedy produces one deterministic team. Random-Search is not included because its refined execution is incomplete.

All 480 outputs of each GA and all sixteen greedy outputs satisfy the recorded mandatory-coverage checks. Of 480 random draws, 417 satisfy those checks; the other 63 are marked as mandatory-coverage failures. Every project's highest-AE random output is feasible, so restricting selection to feasible draws leaves all sixteen representatives unchanged. Table S11 reports selected-team means, not means over the full run distributions. Table S12 gives the selected seeds and project-level scores. Full rosters, all runs, and component scores are in the V2 baseline workbook.

TABLE S11
V2 selected-team mean scores across sixteen projects. Valid counts refer to individual outputs, not selected representatives; all sixteen representatives per method are feasible.

Method	Valid/total	AE	AT	AC
Calibrated/refined GA	480/480	0.59047	0.60866	0.64395
Uniform GA	480/480	0.56755	0.56236	0.66116
Random	417/480	0.44486	0.53612	0.42845
Greedy	16/16	0.46047	0.61730	0.42911

Calibrated GA exceeds uniform GA on fourteen projects and ties on P6 and P11. The paired AE difference has mean 0.02292, median 0.01216, and range 0–0.10713. It exceeds random and greedy on all sixteen projects, with mean differences 0.14561 and 0.13000, respectively. Comparisons use the sixteen projects as units; the thirty seeds contribute to representative selection, not thirty independent project comparisons. Higher refined AE measures alignment with the encoded policy, not stronger practitioner preference. These computational representatives do not replace the teams judged in RQ5.

C. Human-Evaluated LLM Condition and Second Round

The uncontaminated human-evaluated LLM condition yielded first-round judgments of five V1 GA preferences, eight LLM preferences, and three equivalences. It excludes

TABLE S12

V2 PROJECT-LEVEL RESULTS. k: TEAM SIZE; s_G AND s_U: SELECTED CALIBRATED/REFINED- AND UNIFORM-GA SEEDS; f_U: UNIFORM OBJECTIVE USED FOR SELECTION. ALL AT, AC, AND AE COLUMNS USE THE REFINED EVALUATOR AFTER SELECTION. G, U, R, AND D DENOTE CALIBRATED GA, UNIFORM GA, RANDOM, AND DETERMINISTIC GREEDY. EQUAL DISPLAYED VALUES ARE ROUNDED; P6 AND P11 ARE EXACT AE TIES. FULL SELECTED ROSTERS ARE LINKED IN THE TEXT.

Project	k	s_G	s_U	f_U	AE_U	AT_U	AC_U	AE_G	AE_R	AE_D
P1	9	20	3	0.83423	0.61219	0.63147	0.60833	0.62171	0.44775	0.46100
P2	8	24	2	0.80674	0.44591	0.41581	0.59643	0.47125	0.40203	0.41918
P3	5	2	1	0.89593	0.65374	0.64449	0.70000	0.68225	0.46621	0.46681
P4	9	12	7	0.85297	0.65265	0.66589	0.65000	0.65709	0.46201	0.46172
P5	4	4	1	0.91705	0.62574	0.61422	0.68333	0.63073	0.48485	0.52350
P6	7	7	1	0.91272	0.54320	0.52137	0.65238	0.54320	0.43396	0.44007
P7	5	4	7	0.89104	0.61944	0.61733	0.63000	0.63272	0.46462	0.46517
P8	5	2	26	0.90040	0.67514	0.70085	0.67000	0.68348	0.45473	0.48348
P9	6	23	19	0.87577	0.45448	0.41071	0.67333	0.48236	0.42038	0.44359
P10	8	10	1	0.89546	0.61649	0.60908	0.65357	0.62574	0.45076	0.51748
P11	9	14	7	0.85139	0.48230	0.45321	0.62778	0.48230	0.41766	0.42183
P12	4	1	1	0.89583	0.45043	0.40385	0.68333	0.48761	0.36937	0.42415
P13	4	1	1	0.94444	0.73347	0.90079	0.70000	0.74451	0.47914	0.50605
P14	4	3	3	0.93241	0.63737	0.62485	0.70000	0.74451	0.54535	0.49451
P15	5	2	12	0.89444	0.41966	0.36359	0.70000	0.47791	0.40256	0.41026
P16	9	8	27	0.87333	0.45851	0.42021	0.65000	0.48014	0.41640	0.42877

the earlier contaminated nineteen-row condition. Teams were generated through ChatGPT's web interface with the recorded interface model label `GPT-5.6 Sol` and `High` reasoning mode. Three independent executions used the same prompt and input data, incorporating elicited priorities. The predefined selection rule was to retain the first complete, structurally valid response in full, before practitioner judgment. Response 1 was valid and retained; response 2 contained invalid JSON and was discarded in full; response 3 was valid but unused.

In round two, the focal teams were newly generated by the refined calibrated GA. The sixteen LLM teams were reused exactly from response 1; historical rosters were reused from actual project compositions; uniform-GA teams were reused from the earlier baseline execution, without a new search. Blind codes and A/B positions changed alongside the focal teams.

For these practitioner comparisons, refined and uniform GA representatives were selected from thirty seeds per project by each method's own objective, retaining the best valid run and breaking exact ties by the smallest seed. The refined GA used the corrected AE formula after the technology-coverage and collaboration refinements. The uniform GA used its simple-arithmetic-mean objective. Calibrated/refined AE was computed only afterward to rescore uniform teams on a common scale; it did not select their representatives.

The second round contains forty V2 comparisons, without reversed repeats and with GA appearing twenty times in each A/B position. Table S13 preserves the five response categories. "Slight" explicitly states that both teams are acceptable. Historical cases P2, P3, P5, and P10 were omitted because their rosters coincided with LLM rosters. The uniform-GA condition retains the previous twelve-case subset and includes P11 as an identical-composition control judged equivalent; removing it leaves four GA preferences, five equivalences, and two uniform preferences among eleven distinct pairs.

TABLE S13

V2 PREFERENCE COUNTS. G = REFINED GA; C = COMPARATOR; EQ. = NO MEANINGFUL PREFERENCE.

Comparator	G clear	G slight	Eq.	C slight	C clear
LLM	1	7	3	4	1
Historical	0	3	2	4	3
Uniform GA	2	2	6	0	2
Total	3	12	11	8	6

Among twenty-nine decisive judgments, the preferred team covered more SHOULD in eight and tied in twenty-one; it never covered fewer. For COULD, the counts are fourteen higher, fourteen lower, and one equal. No optional open explanations were recorded in this round, so these are indicator associations, not established decision rationales. All twelve historical rosters include a person without a current technology profile; no missing profile data were imputed. Reused cases, one authority, bundled model changes, and a changed response scale prevent interpreting round differences as a controlled treatment effect. The V1 absolute-acceptability rate was not remeasured for V2.

The original-evaluation package and the V2 baseline workbook cover different evidence: the latter releases the refined

computational baseline records, not the raw human-evaluation sessions or LLM exchanges. This supplement keeps those evidence units and model versions separate; access to their numerical summaries does not establish end-to-end reproducibility of every experiment.